Freight Rate Prediction:

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At a glance: predict the rate of 12,000 loads from Nov-Dec 2025 using 48,000 labeled loads from Jan-Oct. Final model: a linear trend layer plus gradient-boosted trees. On the most realistic time-ordered test (Sep-Oct) it is off by about 1.5% on average (median 1.3%).

1. Explore: key findings

- **It is a forecasting task.** Labeled data ends on 31 Oct; the loads to predict are in Nov-Dec.
- **Prices drift upward:** about +6% over Jan-Oct, on top of short-term moves.
- **market_index is mostly a daily factor:** about 98% of its variance is between days (within-day std about 0.025 vs 0.167 between daily means), with a strong 7-day cycle and some level shifts.
- **Structure:** short hauls cost more per mile, and equipment type shifts the price level.

2. Clean: data-quality issues and fixes

Issue	Size	Fix
Corrupted posted_rate (rate/mile outside 1.2-4.0)	675 rows (1.4%)	Dropped from training only
Negative weight (sign flip)	292 rows (0.6%)	Absolute value
Missing weight	300 rows (0.6%)	Median (linear layer); trees handle NaN
Missing market_index	374 rows (0.8%)	Mean of the same day
Noisy distance (e.g. 70 mi on a short lane)	47 rows	Lane median distance

3. Engineer features

Feature	Layer	What it is / why it helps
ld	Trees	Log of the trip distance (miles). Uses the lane's median distance because some raw distances are wrong. Log, so 100 vs 200 miles counts like 500 vs 1,000.
eq	Trees	Equipment type coded as a number (Dry Van, Flatbed, Reefer). Each truck type has its own price level.
w	Trees	Absolute weight of the load. Negative values (sign errors) are made positive; missing values stay empty and trees handle them.
lw	Linear	Log of the weight; missing values replaced by the median.
mi	Trees	market_index of the day. Missing values replaced by that day's mean.
lmi	Linear	Log of market_index, so the model measures the price effect of a % change in the market.
quote_signal	Trees	Raw quote signal from the data. Its business meaning is unknown to me; kept because it improves time-ordered validation.
qs_day	Linear	Average quote_signal of all loads on the same day: the general level of the day.
qs_dev	Linear	quote_signal of this load minus qs_day: how this load differs from the others that day.
t	Linear	Days since 1 Jan 2025. Captures the upward price trend and lets the model extend it into Nov-Dec.
dow	Trees	Day of week (0 = Monday ... 6 = Sunday), for the weekly pattern.
pickup_c, delivery_c	Trees	Pickup and delivery city as category codes. Some cities are more expensive than others.
pickup_lat, pickup_lon, delivery_lat, delivery_lon	Trees	Coordinates of both cities, so the trees can group nearby regions.

- **Target:** log(rate / distance), converted back to dollars by multiplying by the lane distance.
- **Deliberately not used:** month or day of year, because Nov-Dec never appears in training.

4. Validate: time-ordered split

- **Why:** the real task predicts future months, so each test trains on the past and predicts the next two months.
- **Why not random:** market_index is mostly a daily factor (about 98% of its variance is between days), so random folds let the model recognise days it has already seen and look better than it is.

Fold	Train on	Test on	MAPE (clean)	Median APE	Bias (log)
1	Jan-Apr	May-Jun	2.45%	1.74%	+0.007
2	Jan-Jun	Jul-Aug	5.21%	3.61%	-0.023
3 (most realistic)	Jan-Aug	Sep-Oct	1.55%	1.25%	+0.007

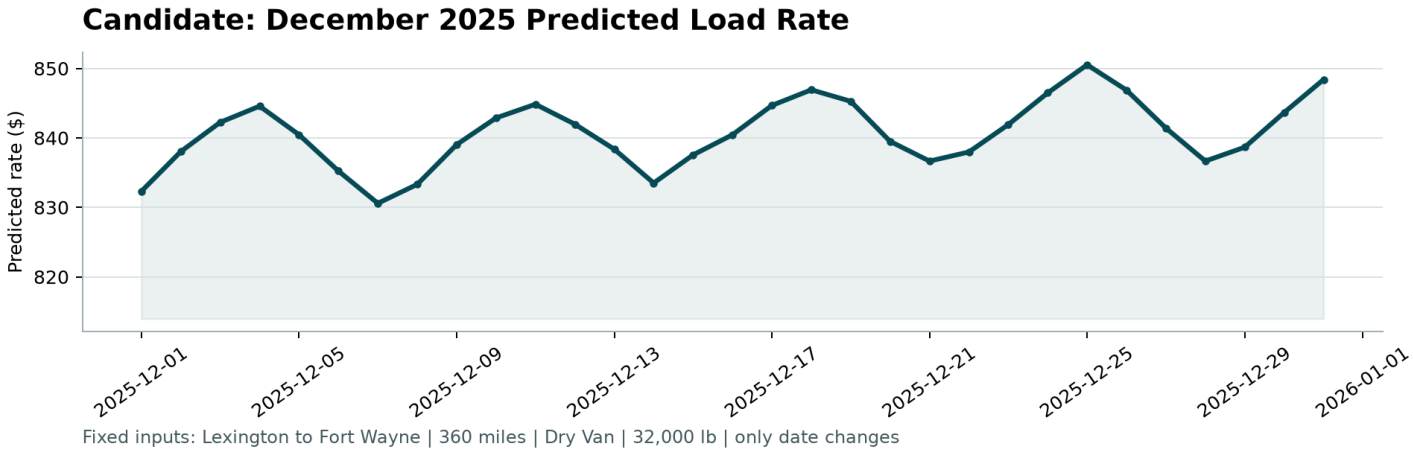
- **Reading it:** metrics use clean rows only (corrupted prices cannot be predicted). Negative bias means over-prediction.
- **Weakest fold (2):** trained only on Jan-Jun, the model over-extrapolated a trend that reversed in July.
- **Final model:** trained on all of Jan-Oct, with corrupted rows removed.

5. Why the hybrid model

Model (mean % error, clean rows)	May-Jun	Jul-Aug	Sep-Oct	Average
Same price per mile for everyone	9.8%	9.0%	9.3%	9.4%
Price per mile by equipment	8.9%	8.0%	8.1%	8.3%
Linear layer only	9.3%	14.0%	9.0%	10.8%
Gradient-boosted trees only (with day count)	3.9%	5.0%	2.1%	3.6%
Hybrid: linear trend + trees (chosen)	**2.4%**	**5.2%**	**1.5%**	**3.1%**

- **Linear layer:** carries the time trend and market sensitivity into months outside the training range.
- **Trees:** learn lane, equipment, distance and quote effects from what the linear layer misses.
- **Evidence:** best in two of three folds and slightly behind trees alone in Jul-Aug (5.2% vs 5.0%). On average 3.1% vs 3.6%, about 0.5 points better.
- **Trade-off:** the hybrid assumes the trend continues; if it reverses (as in July) the advantage disappears.

6. Fixed December chart (produced by score.py)



- **Setup:** same load every day (Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lb); only the date changes.
- **Result:** about $830 to $850 (roughly 2.5% peak to trough). The weekly wave follows market_index; the slow rise follows the learned trend.
- **Caution:** the axis does not start at zero, which exaggerates the swings. The 25 Dec peak is a weekly-cycle effect, not a learned holiday effect.